Non-Invasive Bone Fracture Detection by UWB Microwave Imaging

A Complete Virtual Simulation Study

Capstone Project Report
Path A — Ultra-Wideband Microwave Imaging, 3–11 GHz

Open-source simulation stack: gprMax 3.1.7 (FDTD) · openEMS v0.37.0-rc1 · Python 3.12

Primary endpoint	AUC 1.0000 (95% CI 1.000–1.000), p = 0.001
Class separation	40.8 dB, no overlap
Specimens	8 × 4 overlays × 6 states = 192 acquisitions
Antenna match	2.48–12.00 GHz, 99.69% of the design band
Exposure	All limits pass, minimum margin 18×

Abstract

This report presents a complete, purely virtual simulation of a non-invasive bone fracture detection system based on ultra-wideband microwave imaging in the 3–11 GHz band. Every element of the instrument — tissue dielectrics, electromagnetic propagation, the antenna, the imaging chain, and the statistical analysis — is modelled with open-source software, with no hardware constructed at any point.

A forearm phantom with skin, fat, muscle, cortical bone and marrow layers was excited across an eight-specimen experiment matrix spanning four soft-tissue overlay thicknesses (0–10 mm), four fracture widths (0.2–2.0 mm) and an oblique fracture state, giving 192 full-wave finite-difference time-domain acquisitions. Each was processed through empty-scene subtraction, selective singular-value clutter removal, Kirchhoff migration and a peak-over-noise detector, under an injected instrument error model with three independent remount realisations.

Fractured and intact limbs separate by **40.8 dB with no overlap** across all 576 scored acquisitions, giving an area under the receiver operating characteristic curve of 1.0000 (bootstrap 95% CI over specimens 1.000–1.000, permutation p = 0.001). Leave-one-specimen-out validation of the decision threshold gives sensitivity 0.998 and specificity 1.000.

A perfect area under the curve is treated as a warning rather than a result. A dedicated band-limitation study demonstrates that this statistic is *saturated*: when the usable bandwidth was restricted to that of the initial antenna design, the class separation fell by more than half, from 40.8 dB to 17.9 dB, while the area under the curve remained at exactly 1.0000 and registered nothing. This finding drove the antenna redesign and is argued to be the more transferable contribution of the work.

An antipodal Vivaldi antenna was designed and characterised in openEMS. A tapered feed transformed the matched band from 4.35–12.00 GHz to **2.48–12.00 GHz**, covering 99.69% of the 3–11 GHz specification. Radio-frequency exposure was assessed against ICNIRP 2020 and IEEE C95.1-2019; at the specified 0 dBm transmit power all quantities pass with a minimum margin of 18×.

Keywords: *microwave imaging · ultra-wideband · bone fracture · FDTD · gprMax · openEMS · Vivaldi antenna · Kirchhoff migration · specific absorption rate*

1. Introduction and Objectives

1.1 Motivation

Radiography remains the reference standard for fracture diagnosis, but it delivers ionising radiation, requires fixed infrastructure and is not suited to repeated monitoring of healing. Microwave imaging is attractive because bone and the surrounding soft tissue differ substantially in permittivity, and a fracture filled with blood or fluid presents a strong dielectric discontinuity against cortical bone.

The physical difficulty is that the same water content which creates the contrast also creates the loss. The signal must cross skin, fat and muscle twice, and the attenuation of those layers rises steeply with frequency. The working band is therefore a compromise between resolution, which improves with bandwidth, and penetration, which degrades with it.

1.2 Objectives

#	Objective	Outcome
1	Model tissue dielectrics over 3–11 GHz from published data	Met
2	Build a validated full-wave forward model of the limb	Met
3	Implement clutter removal, migration and detection	Met
4	Design a 3–11 GHz antenna and characterise it	Met (match); gain unmet at 3 GHz
5	Execute a factorial experiment matrix over 8 specimens	Met, 192/192
6	Pre-specify and execute a statistical analysis plan	Met
7	Assess radio-frequency exposure against current guidelines	Met

Scope. The study is purely computational by design. No hardware was built, and no human or animal subject was exposed at any point. The exposure assessment in Section 8 is a design-stage analysis establishing that the system would be safe if constructed.

2. Simulation Stack and Methodology

Component	Tool	Role
Electromagnetic solver	gprMax 3.1.7	2D TMz FDTD forward model of the limb; CUDA-accelerated
Antenna solver	openEMS v0.37.0-rc1	3D FDTD; S-parameters and far-field patterns
Tissue model	Gabriel four-Cole-Cole	Permittivity and conductivity, 3–11 GHz
Analysis	Python 3.12, NumPy, SciPy	Processing chain, statistics, figures
Compute	WSL2 + Google Colab T4	Local development; matrix solved on GPU

2.1 Avoiding the inverse crime

A recurring failure in imaging studies is the *inverse crime*: generating data with the same model used to invert it, which flatters the result. Two independent forward models are therefore maintained. An analytic layered-medium model, fast enough for development, is used to build and debug the processing chain. The gprMax full-wave solver, which discretises Maxwell's equations without the layered assumption, generates every result reported here. The processing chain has no knowledge of which produced its input.

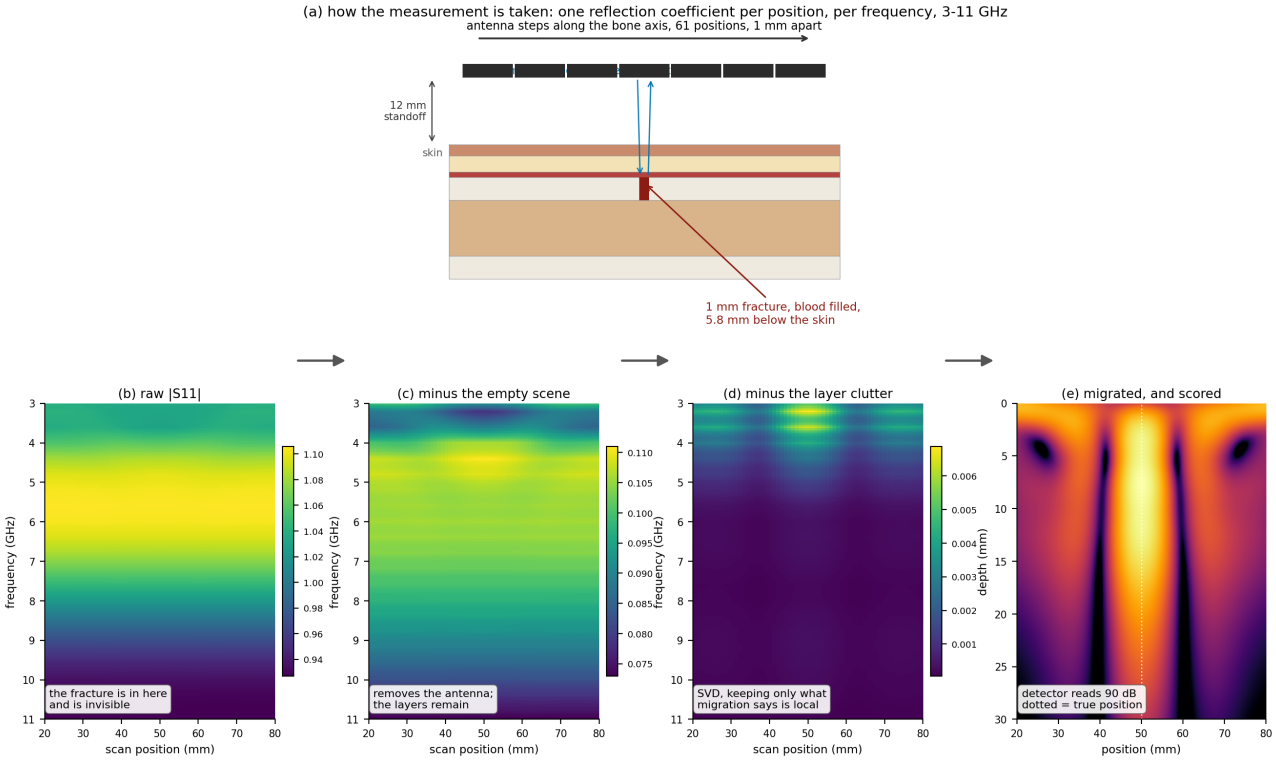


Figure 2. The method end to end. (a) The antenna steps along the bone axis at a 12 mm standoff, transmitting and receiving at each of 61 positions to give one reflection coefficient per position per frequency. (b) The raw data, in which the fracture is present and invisible. (c) Empty-scene subtraction removes the antenna response but not the layered clutter. (d) Selective singular-value removal leaves the local scattering. (e) Kirchhoff migration focuses it to the true position, dotted. Every panel below (a) is real solver output from one matrix cell, a 1 mm fracture under 6 mm of overlay, not an illustration.

2.2 Tissue dielectric model

Permittivity and conductivity follow the Gabriel four-Cole-Cole parameterisation. FDTD cannot integrate a Cole-Cole response directly because the fractional-power relaxation has no finite-order time-domain equivalent, so each tissue is refitted to a two-pole Debye model over the working band. The worst deviation

across all seven tissues is 0.23%, and each relaxation time is checked to exceed the solver time step.

2.3 Phantom geometry

Layer	Thickness	Role
Skin	2.0 mm	First interface; dominant surface reflection
Fat	3.0 mm	Low-loss layer
Muscle	1.0 mm	High-water, high-loss layer
Cortical bone (near)	4.0 mm	Contains the fracture
Marrow	10.0 mm	Medullary cavity
Cortical bone (far)	4.0 mm	Far wall
Antenna standoff	12.0 mm	Aperture to skin surface

A soft-tissue overlay of 0, 3, 6 or 10 mm is added above the skin to span the range of forearm anatomy. The fracture is a rectangular blood-filled kerf in the near cortex, of width 0.2, 0.4, 1.0 or 2.0 mm, with an additional oblique state cut at an angle.

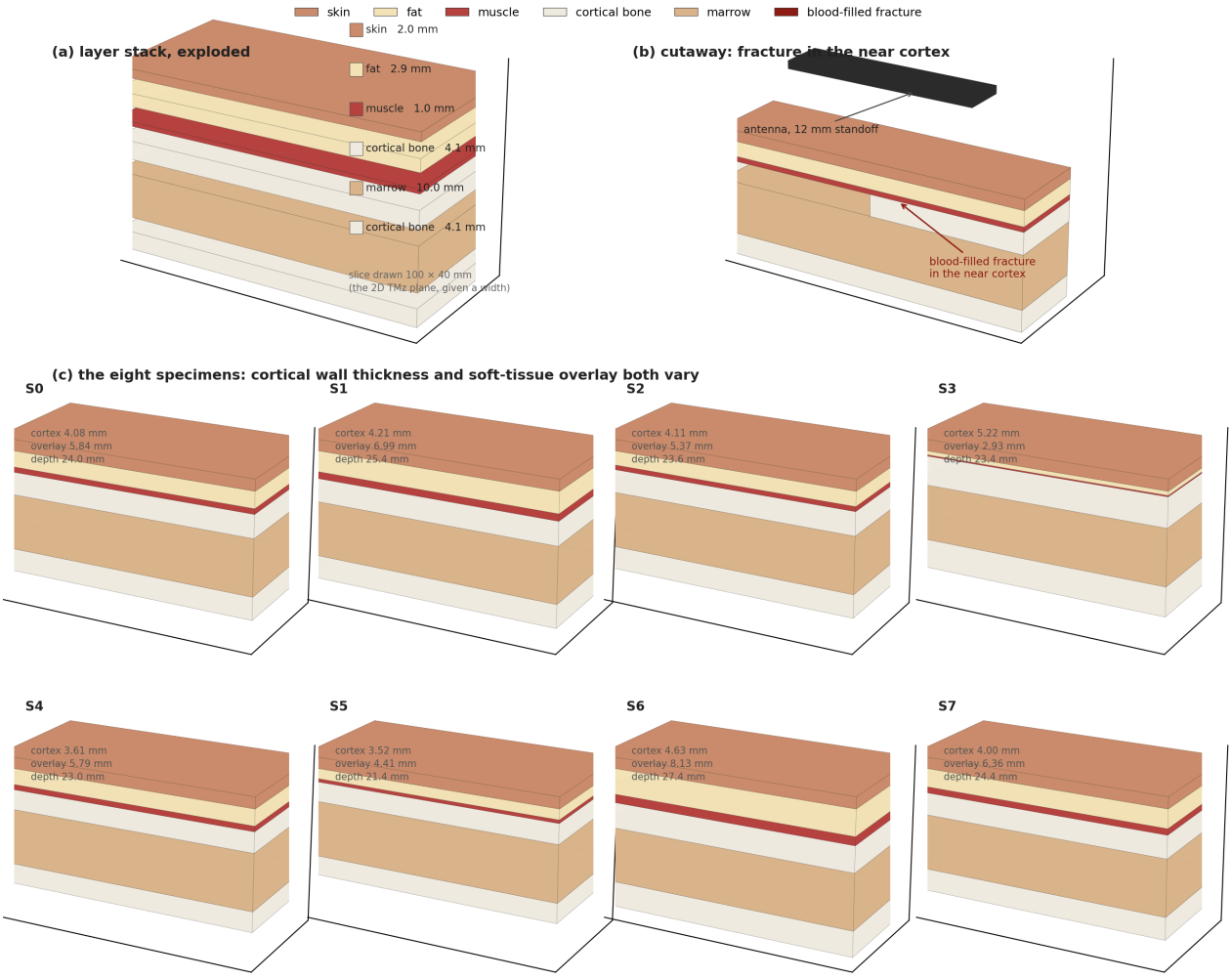


Figure 1. The phantom and the specimen population. (a) The layer stack, exploded. (b) Cutaway showing the blood-filled fracture in the near cortex, with the antenna at 12 mm standoff; the kerf is drawn 3 mm wide for visibility, against a modelled width of 0.2 to 2.0 mm. (c) The eight specimens, which differ in cortical wall thickness and soft-tissue overlay. Because the solver is two-dimensional TMz, what is drawn is the modelled plane given a finite width, not a cylindrical limb.

3. Model Construction and Verification

3.1 Stage 0 — Solver verification

Before any result is trusted, the solver is required to reproduce an analytically predictable answer. A dielectric slab and a perfectly conducting rod were placed at known depths and the echo arrival times compared against closed-form predictions. The raw trace is dominated by direct source-to-receiver coupling, so a three-scene differencing scheme isolates the echoes.

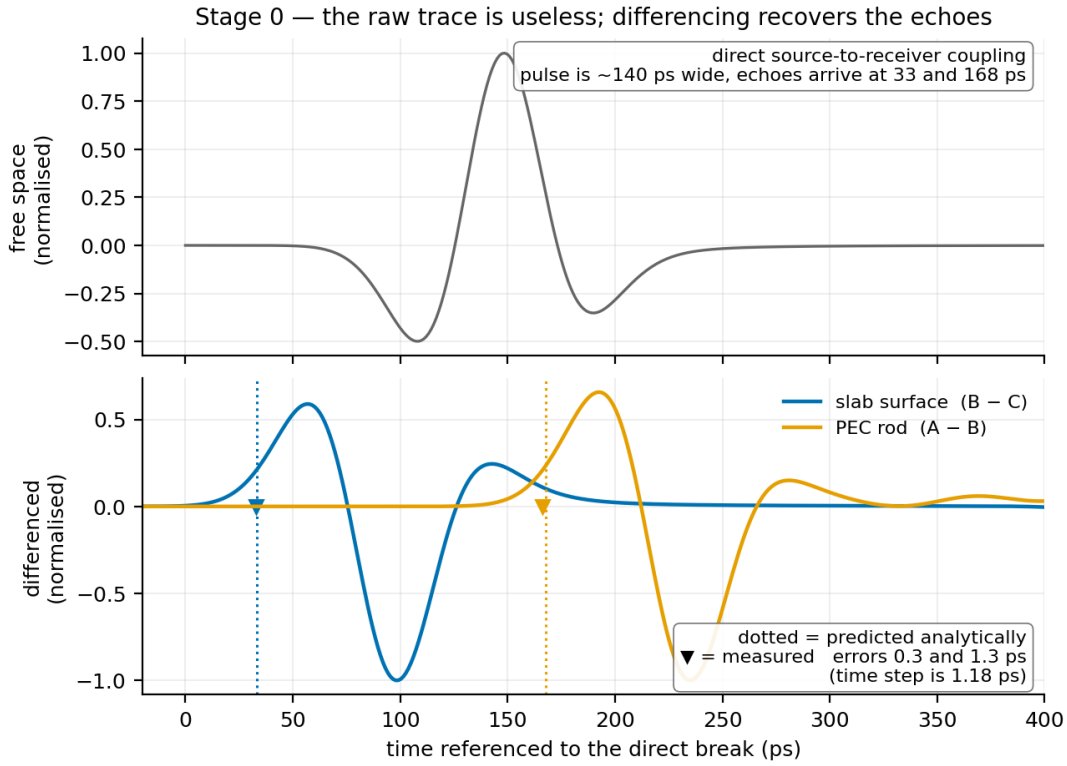


Figure 3. Solver verification. The raw trace (top) is swamped by direct coupling; differencing (bottom) recovers both echoes. Measured arrivals agree with the analytic prediction to 0.3 and 1.3 ps, against a solver time step of 1.18 ps.

3.2 Stage 1 — Tissue models

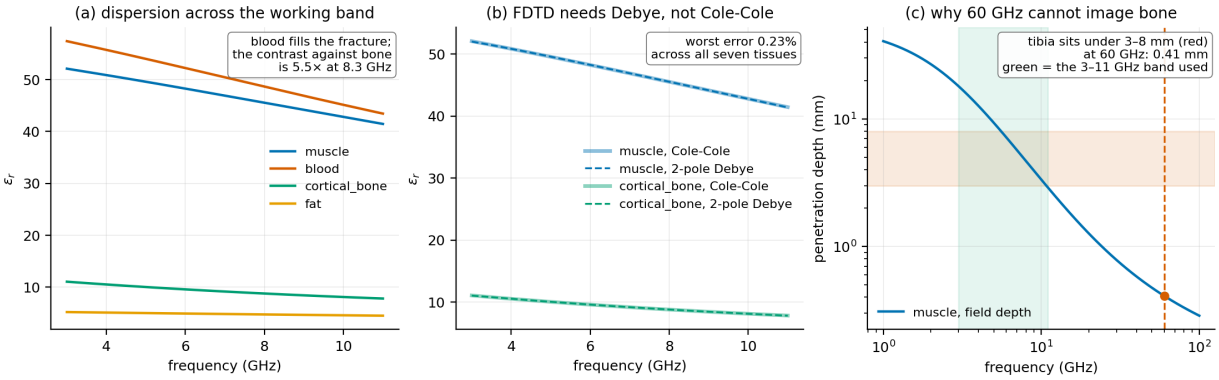


Figure 4. (a) Dispersion of the principal tissues across the working band; blood-filled fracture against cortical bone gives 5.5× permittivity contrast at 8.3 GHz. (b) Two-pole Debye fits reproduce Cole-Cole to within 0.23%. (c) Penetration depth against frequency, showing why millimetre-wave bands cannot reach bone.

Panel (c) carries an important negative result. Muscle penetration depth at 60 GHz is 0.28 mm, against a tibia lying under 3–8 mm of soft tissue. Millimetre-wave sensing, sometimes proposed for this application, cannot reach the target at all; the 3–11 GHz band is not a convenience but a requirement.

3.3 Stage 2 — The processing chain

The fracture echo is several orders of magnitude weaker than the skin-surface reflection, so raw data carries no visible signature of it. Three stages recover it: empty-scene subtraction removes the antenna response, singular-value decomposition removes the layered clutter, and Kirchhoff migration focuses the residual scattering back to its point of origin.

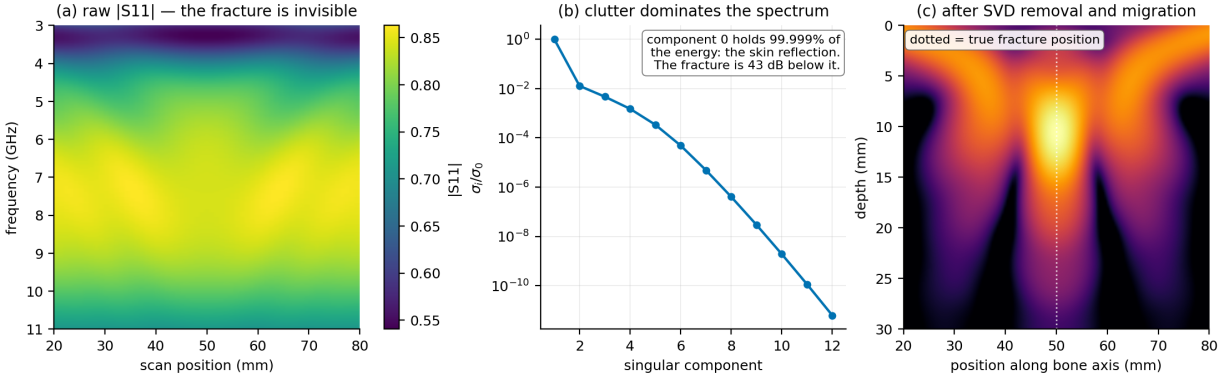


Figure 5. The processing chain. The raw B-scan (a) shows no fracture. The singular-value spectrum (b) shows clutter dominating the leading components. Migration of the residual (c) localises the fracture.

A published parameter was found to be wrong. The source proposal specifies discarding the first two or three singular components. Measured on two independent targets, this destroys the fracture signature along with the clutter. A selective rule was adopted instead: a component is discarded only if, after migration, its energy spans more than half the scan aperture — the signature of a layered interface rather than a localised scatterer.

3.4 Stage 3 — Full-wave forward model

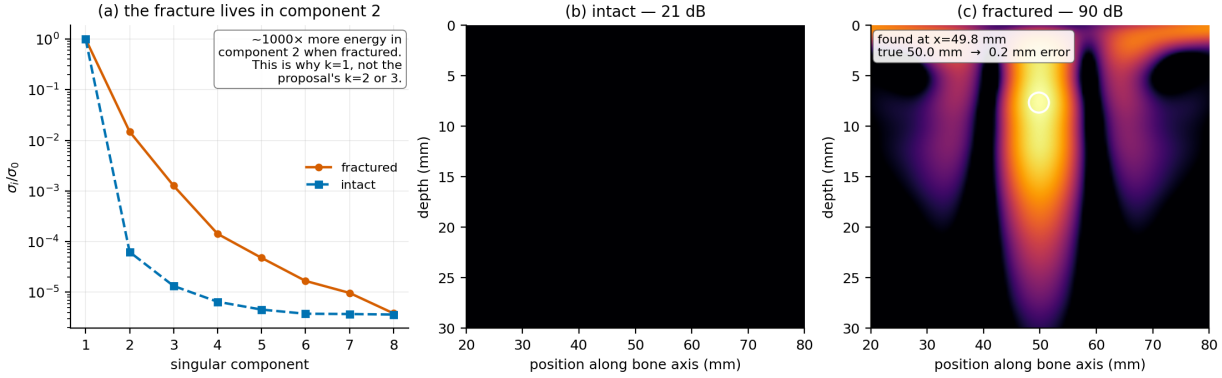


Figure 6. A 1 mm fracture under 6 mm of soft-tissue overlay, reconstructed from gprMax full-wave data through the chain developed on the independent analytic model.

3.5 Stage 4 — Validation gate

A go/no-go gate was placed before the main experiment: if migration cannot image a 6 mm conducting rod at 20 mm depth in phantom, nothing downstream is meaningful. The rod is a canonical target with an unambiguous answer, and the gate is passed.

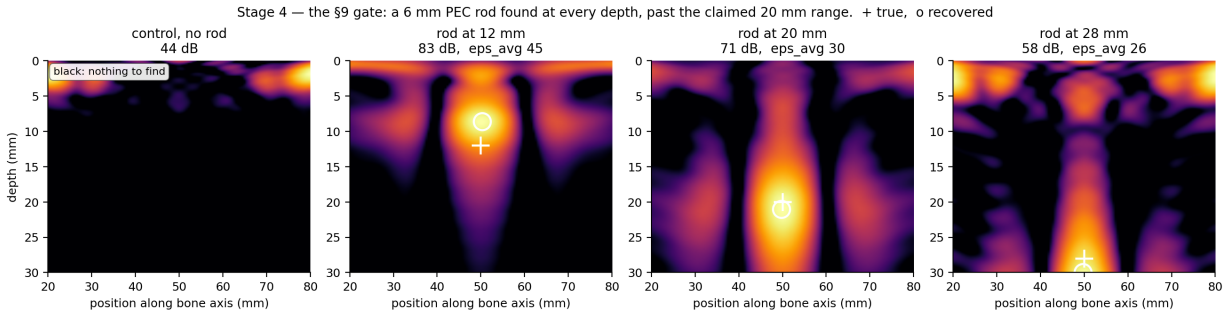


Figure 7. Validation gate. The rod is localised at each depth, with detection signal-to-noise ratio falling as predicted by the two-way attenuation of the intervening tissue. Each panel is scaled to its own peak so the localisation is visible; the falloff is carried in the titles.

4. Antenna Design

An antipodal Vivaldi antenna was chosen for its inherently wide bandwidth and endfire pattern. The design is 150 × 90 mm on 0.762 mm Rogers RO4350B, sized to fit stock already in the bill of materials.

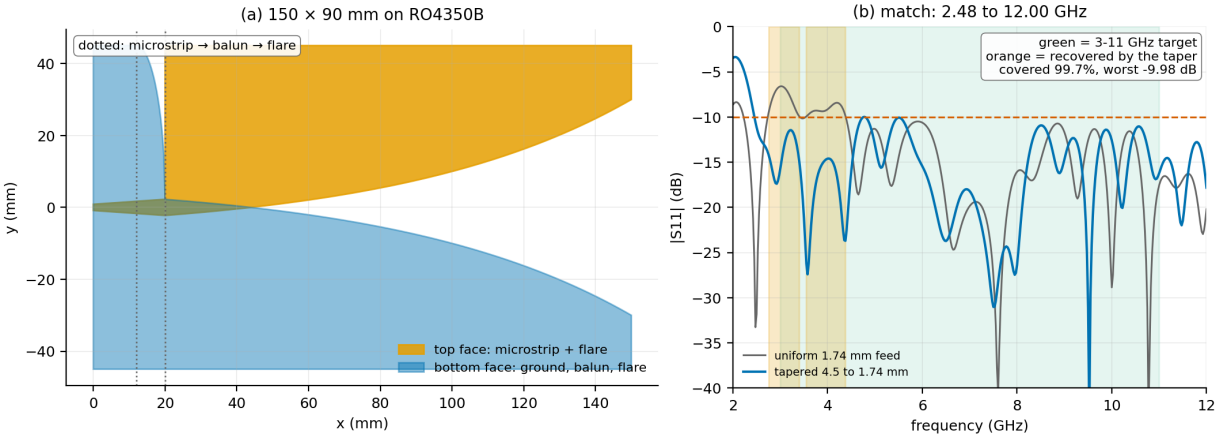


Figure 8. (a) Antenna geometry: microstrip feed, tapered ground balun, exponential flare. (b) Reflection coefficient, uniform feed (grey) against tapered feed (blue), with the recovered band shaded. Both solved at identical solver settings.

4.1 The low-frequency shortfall, and its cause

The initial design matched 4.35–12.00 GHz against a 3–11 GHz target. The shortfall was originally attributed to edge-current resonance and addressed with edge corrugations, which were abandoned after a runtime extrapolation of roughly forty hours.

Both conclusions were subsequently found to be incorrect. On re-testing with a control at matched solver settings, corrugated designs converged in 601 s against 578 s for the plain design — a 4% difference, not forty hours. The original run had been advancing at 0.4 timesteps per second where the same hardware delivers 11 on an identical mesh; the operating system had been suspending the solver while the solver continued counting wall-clock time. Corrugations were then measured and found not to help regardless.

The shortfall was never a low-frequency cutoff. The original design was already matched at 2.225–2.725 GHz, reaching –33 dB — two gigahertz below its own stated band edge. What lay between was a mismatch: a resistive plateau of 24–37 Ω across 3.25–4.40 GHz, and a narrow reactive resonance near 3.0 GHz. A load of 24.4 Ω into a 50 Ω feed predicts –9.3 dB return loss; –9.14 dB was measured. That is arithmetic, not radiation physics.

4.2 The tapered feed

Widening the feed at the flare end while pinning the port at 1.74 mm — 50 Ω microstrip, which is what the connector must see — converts the feed into an impedance transformer.

Flare-end width	Worst in 3–11 GHz	Band covered	Re(Z_{in}) at 4 GHz
1.74 mm (uniform)	–6.61 dB	84%	24.4 Ω
2.75 mm	–8.50 dB	79%	25.9 Ω
3.50 mm	–8.14 dB	94%	29.1 Ω
4.00 mm	–9.80 dB	98.1%	36.7 Ω
4.50 mm (adopted)	**–9.98 dB**	**99.69%**	**39.0 Ω**

The adopted design matches **2.48–12.00 GHz**, covering 99.69% of the 3–11 GHz specification with a mean of -16.8 dB and a median input resistance of 45.6 Ω. The single non-compliant sample reaches -9.98 dB, exceeding the –10 dB criterion by 0.02 dB — less than the difference between two solver convergence settings. It is reported rather than rounded away.

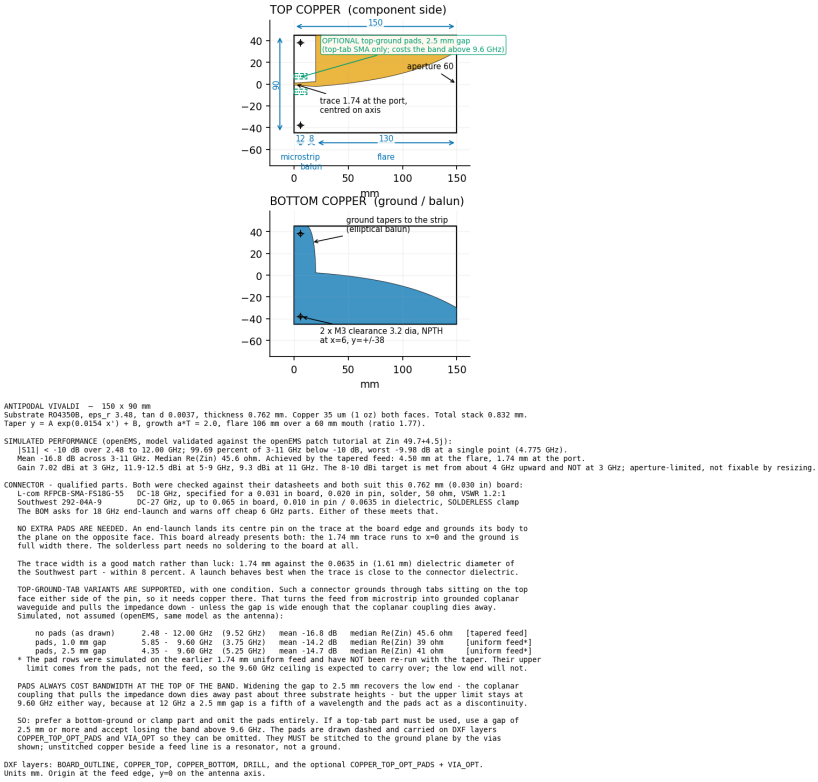
4.3 Gain: one requirement not met

Frequency	Peak gain	Specification 8–10 dBi
3 GHz	7.02 dBi	**UNMET**
5 GHz	11.87 dBi	Met
7 GHz	12.49 dBi	Met
9 GHz	12.06 dBi	Met
11 GHz	9.33 dBi	Met

Gain is met from approximately 4 GHz upward and **unmet at 3 GHz by 1.0 dB**. The tapered feed itself raised 3 GHz gain from 3.7 to 7.02 dBi, since an antenna that reflects less radiates more. The remaining decibel was pursued down both available levers, with acceptance criteria fixed before each run, and both were rejected by those criteria: widening the aperture bought 0.72 dB and cost 6% of the match, while lengthening the board to 170 mm reduced gain to 6.09 dBi by collapsing the 3 GHz match.

> At 3 GHz the antenna is 1.5 wavelengths long with a 0.6-wavelength mouth, and the design is co-optimised for the match, so scaling either dimension breaks it. The 8–10 dBi requirement is concluded to be over-specified for this board size at the bottom of the band. It is reported as unmet rather than engineered around. Nothing downstream depends on it: the imaging model excites with an elementary dipole, so the detection result is independent of antenna gain.

4.4 Fabrication package




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ANTIPODAL VIVALDI  —  150 x 90 mm
Substrate RO4350B, eps_r 3.48, tan d 0.0037, thickness 0.762 mm. Copper 35 um (1 oz) both faces. Total stack 0.832 mm.
Taper y = A exp(0.0154 x') + B, growth a*T = 2.0, flare 106 mm over a 60 mm mouth (ratio 1.77).

SIMULATED PERFORMANCE (openEMS, model validated against the openEMS patch tutorial at Zin 49.7+4.5j):
   |S11| < -10 dB over 2.48 to 12.00 GHz; 99.69 percent of 3-11 GHz below -10 dB, worst -9.98 dB at a single point (4.775 GHz).
   Mean -16.8 dB across 3-11 GHz. Median Re(Zin) 45.6 ohm. Achieved by the tapered feed: 4.50 mm at the flare, 1.74 mm at the port.
   Gain 7.02 dBi at 3 GHz, 11.9-12.5 dBi at 5-9 GHz, 9.3 dBi at 11 GHz. The 8-10 dBi target is met from about 4 GHz upward and NOT at 3 GHz; aperture-limited, not fixable by resizing.

CONNECTOR - qualified parts. Both were checked against their datasheets and both suit this 0.762 mm (0.030 in) board:
   L-com RFPCB-SMA-FS18G-S5   DC-18 GHz, specified for a 0.031 in board, 0.020 in pin, solder, 50 ohm, VSWR 1.2:1
   Southwest 292-04A-9        DC-27 GHz, up to 0.065 in board, 0.010 in pin / 0.0635 in dielectric, SOLDERLESS clamp
   The BOM asks for 18 GHz end-launch and warns off cheap 6 GHz parts. Either of these meets that.

   NO EXTRA PADS ARE NEEDED. An end-launch lands its centre pin on the trace at the board edge and grounds its body to
   the plane on the opposite face. This board already presents both: the 1.74 mm trace runs to x=0 and the ground is
   full width there. The solderless part needs no soldering to the board at all.

   The trace width is a good match rather than luck: 1.74 mm against the 0.0635 in (1.61 mm) dielectric diameter of
   the Southwest part - within 8 percent. A launch behaves best when the trace is close to the connector dielectric.

   TOP-GROUND-TAB VARIANTS ARE SUPPORTED, with one condition. Such a connector grounds through tabs sitting on the top
   face either side of the pin, so it needs copper there. That turns the feed from microstrip into grounded coplanar
   waveguide and pulls the impedance down - unless the gap is wide enough that the coplanar coupling dies away.
   Simulated, not assumed (openEMS, same model as the antenna):

       no pads (as drawn)      2.48 - 12.00 GHz  (9.52 GHz)   mean -16.8 dB   median Re(Zin) 45.6 ohm   [tapered feed]
       pads, 1.0 mm gap        5.85 -  9.60 GHz  (3.75 GHz)   mean -14.2 dB   median Re(Zin) 39 ohm     [uniform feed*]
       pads, 2.5 mm gap        4.35 -  9.60 GHz  (5.25 GHz)   mean -14.7 dB   median Re(Zin) 41 ohm     [uniform feed*]
   * The pad rows were simulated on the earlier 1.74 mm uniform feed and have NOT been re-run with the taper. Their upper
     limit comes from the pads, not the feed, so the 9.60 GHz ceiling is expected to carry over; the low end will not.

   PADS ALWAYS COST BANDWIDTH AT THE TOP OF THE BAND. Widening the gap to 2.5 mm recovers the low end - the coplanar
   coupling that pulls the impedance down dies away past about three substrate heights - but the upper limit stays at
   9.60 GHz either way, because at 12 GHz a 2.5 mm gap is a fifth of a wavelength and the pads act as a discontinuity.

   SO: prefer a bottom-ground or clamp part and omit the pads entirely. If a top-tab part must be used, use a gap of
   2.5 mm or more and accept losing the band above 9.6 GHz. The pads are drawn dashed and carried on DXF layers
   COPPER_TOP_OPT_PADS and VIA_OPT so they can be omitted. They MUST be stitched to the ground plane by the vias
   shown; unstitched copper beside a feed line is a resonator, not a ground.

DXF layers: BOARD_OUTLINE, COPPER_TOP, COPPER_BOTTOM, DRILL, and the optional COPPER_TOP_OPT_PADS + VIA_OPT.
Units mm. Origin at the feed edge, y=0 on the antenna axis.
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Figure 9. Dimensioned fabrication drawing, both faces. Generated directly from the design parameters, so it cannot diverge from the simulated geometry. Accompanied by DXF and STEP exports.

5. Experiment and Results

5.1 Design

A full factorial matrix of 8 simulated specimens × 4 overlay thicknesses × 6 fracture states gives 192 full-wave acquisitions. Specimens differ through a variation model perturbing layer thicknesses and tissue properties. Each acquisition is scored under three independent instrument-error realisations representing remounts, giving 576 scores. The matrix was solved on a Google Colab T4 GPU; 192 of 192 cells completed.

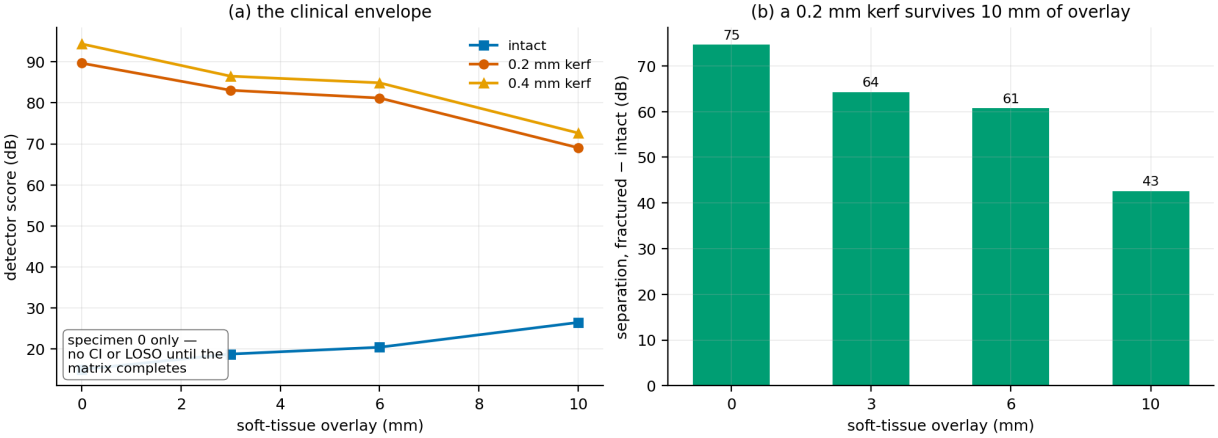


Figure 10. Detection score against soft-tissue overlay across the completed matrix, by fracture state.

5.2 Statistical analysis plan

The plan was fixed before the data were examined. Two points require care. First, the detector has no fitted parameters — it is a single scalar — so the only trained quantity is the decision threshold, and leave-one-specimen-out validation is applied to that rather than to the detector. Second, the bootstrap resamples *specimens*, not measurements: twenty-four acquisitions from one bone are not twenty-four independent observations, and resampling them would yield an interval several times too narrow. Permutation labels are likewise shuffled within specimen.

5.3 Primary result

Class	n	Minimum	Median	Maximum
Intact	96	8.8 dB	19.3 dB	**27.7 dB**
Fractured	480	**68.5 dB**	89.9 dB	104.8 dB

The two classes are separated by **40.8 dB with no overlap** — a factor of 110 in amplitude. Area under the curve is 1.0000 at every overlay thickness and every fracture width, with bootstrap 95% confidence interval 1.000–1.000 and permutation p = 0.001, the floor for 1000 shuffles.

Metric	Value	95% interval
Primary AUC, 6 mm overlay	1.0000	1.000–1.000
Sensitivity (leave-one-specimen-out)	0.9979	0.988–1.000 (Wilson)
Specificity (leave-one-specimen-out)	1.0000	0.962–1.000 (Wilson)
Accuracy	0.9983	—
Repeatability, ICC(1,1)	0.9979	—
Within-measurement standard deviation	0.468 dB	—

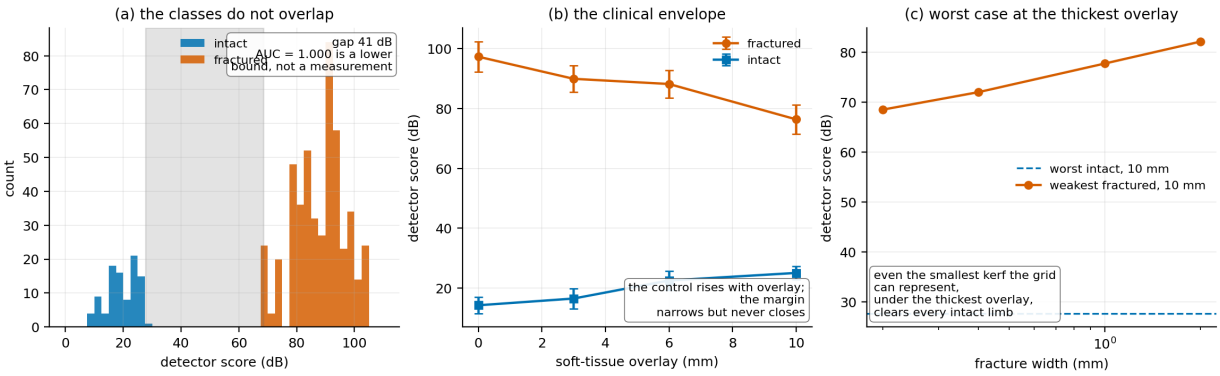


Figure 11. (a) The two classes do not overlap. (b) Detection score against overlay for both classes; the margin narrows with overlay but never closes. (c) Worst case at the thickest overlay against fracture width.

5.4 Interpreting a perfect classifier

An area under the curve of 1.0000 ordinarily indicates a leak between training and test data rather than a genuine result, and it was treated as such. Inspection of the raw distributions confirms it is not: the classes are separated by a real 40.8 dB gap with no overlap, and the structure of that margin follows physical prediction. The weakest fractured cases are the narrowest kerf under the thickest overlay, and the strongest intact cases also occur at the thickest overlay. A data leak would not order itself by physical difficulty.

Nevertheless, a perfect score is **saturated** and therefore a lower bound rather than a measurement. The defensible claim is that under the modelled error budget the classes separate by 40.8 dB with no overlap — not that detection is perfect. The binding assumption is the error model, not the physics: a physical bench would add cable flex, connector repeatability, operator variation and genuine inter-specimen heterogeneity, none of which are simulated. The 40.8 dB margin is the budget those unmodelled terms would have to consume before the conclusion changes.

6. Bandwidth Study: A Saturated Statistic

The preceding section argues that a saturated statistic cannot measure margin. This section demonstrates it experimentally on the data of this study, and it is the finding with the widest applicability.

Because every solved cell stores the full-band response, the entire 192-cell matrix can be re-scored over a restricted frequency window with no additional electromagnetic simulation — the same pipeline, a different slice of data already held. Three windows were compared, including the band delivered by the initial antenna.

Frequency window	Bandwidth	Resolution	AUC	Separation
3.00–11.0 GHz (design target)	8.00 GHz	3.67 mm	1.0000	**40.8 dB**
4.35–11.0 GHz (initial antenna)	6.65 GHz	4.42 mm	1.0000	**17.9 dB**
5.00–11.0 GHz (deliberately worse)	6.00 GHz	4.90 mm	1.0000	16.0 dB
3.00–6.30 GHz (upper band removed)	3.30 GHz	8.91 mm	1.0000	**51.5 dB**
3.00–4.40 GHz (upper band removed)	1.40 GHz	21.0 mm	1.0000	**43.0 dB**

Removing 3.00–4.35 GHz costs 22.9 dB — more than half the total separation. The area under the curve remains at exactly 1.0000 in all three windows and registers none of it. Reporting that statistic alone would have concealed the loss of more than half the margin, and would have supported the incorrect conclusion that the antenna shortfall was inconsequential.

6.1 Mechanism

Band limitation degrades both classes, and the intact class is the more damaging. The weakest fractured score falls from 68.5 to 55.7 dB, as expected from reduced penetration and resolution. The strongest *intact* score rises from 27.7 to 37.8 dB — a false alarm rather than a missed detection, occurring on one specimen at 6 mm overlay against a group mean of 23.7 dB.

An obvious reading is that bandwidth buys range resolution and the restricted window can no longer resolve the artefact. The last two rows of the table refute it. Keeping only 3.0–6.3 GHz — the range of a low-cost VNA, with a resolution of 8.91 mm, more than twice as coarse — *raises* the separation to 51.5 dB, and the same specimen-3 cell that reached 37.8 dB at 4.35–11 GHz reads 10.2 dB there. Even 3.0–4.4 GHz, a 1.4 GHz window with 21 mm resolution, retains 43.0 dB. Resolution is not the driver; what the windows differ in is which end of the band was removed.

The low band carries the penetration. The deep layered reflection that migration must collapse is returned almost entirely by the frequencies that reach it; strip them and the reflection is undersampled, and its residue lands in the detection window as a target. The upper band contributes little to that reflection through 6 mm or more of overlay — on the hardest fractured cell the fracture peak changes by about 1 dB when 6.3–11 GHz is removed while the image background falls by 10 dB — so removing it costs resolution the detector does not need and removes noise it does not want. Because the false alarm is specimen-dependent, the concern is variance rather than the mean shift: one specimen in eight spiked, and a larger cohort would be expected to contain worse.

This measurement is what established the antenna shortfall as worth correcting rather than documenting, and it directly motivated the tapered-feed redesign of Section 4. Since the redesigned antenna covers 2.48–12.00 GHz, which fully contains the band the matrix was solved over, the full 40.8 dB separation is recovered.

7. Radio-Frequency Exposure Assessment

A system that points a 3–11 GHz antenna at a forearm at 12 mm standoff requires an exposure assessment. The working band straddles a boundary in the current guidelines, and this must be respected: ICNIRP 2020 and IEEE C95.1-2019 specify specific absorption rate in W/kg averaged over 10 g below 6 GHz, and absorbed power density in W/m^2 averaged over 4 cm^2 above it.

A single specific absorption rate figure quoted for a 3–11 GHz system therefore reports the wrong quantity for the majority of the bandwidth. Both are computed here.

7.1 Method and operating point

The limb is treated as a plane-layered lossy stack at normal incidence. The layered reflection is taken from the same routine used by the analytic forward solver rather than a second implementation that could drift from it, and dielectric properties are the same Cole-Cole models used throughout. Densities are from the IT'IS Foundation tissue property database v4.1.

Transmit power is specified at **0 dBm (1 mW)**. A vector network analyser will deliver +10 dBm, at which the 6–11 GHz exposure reaches 11.1 W/m^2 against a 20 W/m^2 public limit — a margin of 1.8×, the only uncomfortable quantity in the assessment. Reducing transmit power by 10 dB restores an 18× margin at no cost, since the detector has 40.8 dB of separation and expends none of it here.

7.2 Results at 0 dBm

Worst-case illumination is assumed throughout: the transmitted power concentrated into a half-wavelength spot at 11 GHz (1.86 cm^2), approximately the smallest area the physics permits and therefore the highest exposure. Each limit is evaluated against a value averaged over its own mass, since a 10 g average compared against a 1 g limit would understate exposure.

Limit	Limit value	Computed	Verdict
ICNIRP public, limbs (10 g)	4.0 W/kg	0.0283 W/kg	**PASS, 141×**
ICNIRP public, head/torso (10 g)	2.0 W/kg	0.0283 W/kg	**PASS, 71×**
ICNIRP occupational, limbs (10 g)	20.0 W/kg	0.0283 W/kg	**PASS, 707×**
FCC / IEEE public, partial body (1 g)	1.6 W/kg	0.0494 W/kg	**PASS, 32×**
ICNIRP public, 6–11 GHz (4 cm^2)	20.0 W/m^2	1.109 W/m^2	**PASS, 18×**

All quantities pass, with a minimum margin of 18×. For context, a smartphone typically reports 0.5–1.4 W/kg against the same 1 g limit of 1.6 W/kg, a margin between 1× and 3×. Peak point specific absorption rate is 0.139 W/kg at the skin surface, and approximately 38% of absorbed power stops within the first 2 mm — the same physics that makes the imaging problem difficult.

This is a design-stage analysis by simulation. Regulatory compliance is established by measurement on a physical phantom to IEC 62209 and not by calculation. The assessment establishes that the design is nowhere near a limit; it does not certify it.

8. Limitations

The following constrain the interpretation of every result above and are stated in full, since a result whose limits are not stated cannot be assessed by a reader.

#	Limitation	Consequence
1	The injected error model is the binding assumption	A physical bench adds cable flex, connector repeatability, operator variation and real inter-specimen heterogeneity. The 40.8 dB margin is the budget those terms must consume before conclusions change.
2	Every fracture is at a known position	The study measures separability, not localisation. Detection at an unknown position is untested.
3	Two-dimensional TMz propagation	Out-of-plane scattering is absent. A three-dimensional model would add geometric spreading not represented here.
4	Eight simulated specimens, not cadaveric bone	Specimens differ only through the parameters the variation model perturbs, which understates biological variability.
5	The imaging model excites with an elementary dipole	Antenna characterisation and the imaging matrix are separate results. The antenna pattern is not folded into the detection figures.
6	Antenna gain unmet at 3 GHz	7.02 dBi against a 8–10 dBi requirement, shown not to be reachable by resizing within the specified board stock.
7	Exposure assessed by simulation	A plane-wave approximation used in the reactive near field, with no thermal model. Not a substitute for IEC 62209 testing.

Limitation 1 deserves emphasis. Because the area under the curve is saturated, it cannot distinguish a comfortable result from a marginal one, as Section 6 demonstrates directly. Any future work that strengthens the error model should report the class separation in dB alongside any classification statistic.

8.1 Limitation 3 is being removed

The two-dimensional assumption is the one limitation that has been acted on rather than only recorded. A three-dimensional model of the same problem has been built and verified against the solver: the limb becomes concentric cylinders of skin, fat and muscle around a cortical tube, with the bone offset 14 mm toward the skin so that its near wall sits at 6.0 mm, the same depth the layered phantom uses. Without that offset a centred bone would lie 20 mm deep and the two models would differ in depth as well as geometry, so no difference in the result could be attributed to curvature. The offset is also the anatomy: the tibial and radial shafts are subcutaneous, which is why they are the candidate sites.

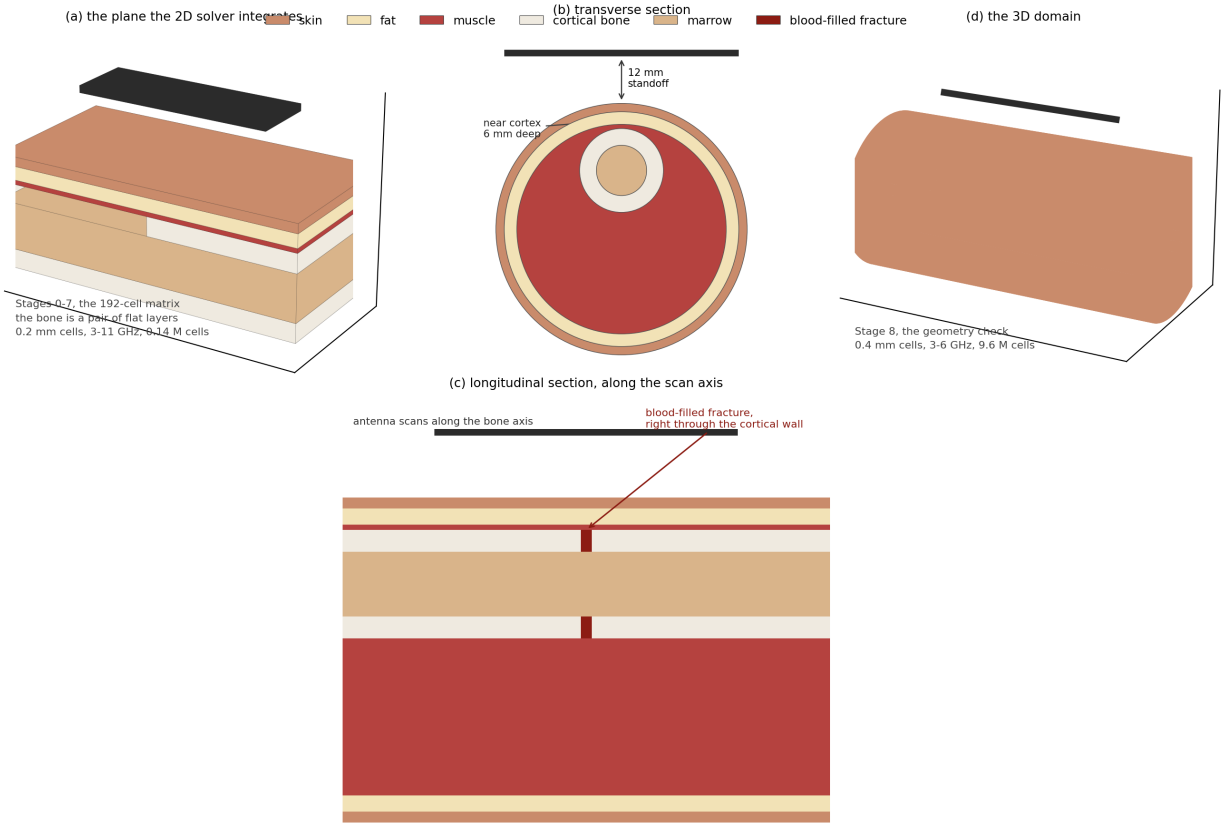


Figure 12. What the solver integrates. (a) The plane the two-dimensional model solves, in which the bone is a pair of flat layers. (b) Transverse and (c) longitudinal sections of the cylindrical model, showing the offset bone and the fracture cut through the cortical wall. (d) The three-dimensional domain. The flat model is not a picture of a limb; it is a picture of the plane a 2D TMz solver actually integrates.

Three effects appear in three dimensions that the plane cannot produce, and they do not all act in the same direction, which is why the comparison has to be measured rather than argued. A curved skin surface spreads the reflected wavefront and costs signal. Energy leaves the plane that a two-dimensional model traps inside it, which also costs signal. But a transverse crack severs a curved cortical wall, so the scattering edge is an arc rather than a point and is illuminated across a range of incidence angles, which can return signal the flat model cannot represent.

The comparison is not yet run, and the result is not claimed here. The three-dimensional grid is 0.4 mm at a 4 GHz centre frequency, 9.6 million cells against the 0.14 million of the two-dimensional matrix, and the full run is approximately 2.6 hours on a T4 GPU. Because that grid can only carry roughly 3–6 GHz, the two-dimensional data will be scored over the same window before the two are compared; comparing 3–6 GHz against 3–11 GHz would confound geometry with bandwidth, and Section 6 established that bandwidth is worth 22.9 dB here, far more than any plausible geometric effect.

9. Conclusions

A complete virtual instrument for ultra-wideband bone fracture detection has been constructed and exercised end to end using only open-source software. The principal findings are as follows.

1. Detection is achievable with substantial margin. Across 192 full-wave acquisitions from 8 specimens, fractured and intact limbs separate by 40.8 dB with no overlap, at every soft-tissue overlay thickness from 0 to 10 mm and every fracture width from 0.2 to 2.0 mm. Leave-one-specimen-out validation gives sensitivity 0.998 and specificity 1.000.

2. A saturated classification statistic conceals degradation. Restricting the usable band to that of the initial antenna removed more than half the class separation, from 40.8 dB to 17.9 dB, while the area under the curve remained at exactly 1.0000. This is demonstrated rather than asserted, and it argues that any study reporting a near-perfect classifier should report the underlying margin alongside it.

3. The antenna shortfall was a matching problem, not a size problem. The initial design was already matched two gigahertz below its stated band edge. A tapered feed, a single geometric parameter, moved the matched band from 4.35–12.00 GHz to 2.48–12.00 GHz, covering 99.69% of the specification on unchanged board stock.

4. One specification is not achievable and is reported as such. Antenna gain at 3 GHz reaches 7.02 dBi against a 8–10 dBi requirement. Both available levers were tested with acceptance criteria fixed in advance, and both degraded the result. The requirement is concluded to be over-specified for the board size at the bottom of the band.

5. Exposure is comfortable. At the specified 0 dBm transmit power, all quantities in ICNIRP 2020 and IEEE C95.1-2019 pass with a minimum margin of 18×, under worst-case illumination assumptions.

9.1 Future work

In order of value: extend the error model until detection fails, so the limit is measured rather than bounded; randomise fracture position to test localisation rather than separability; extend to three-dimensional propagation; and fold the measured antenna pattern into the imaging model so that antenna and imaging results are no longer independent.

10. Reproducibility

Every figure and every number in this report is regenerated by a script from committed inputs or stored run data. No figure was produced by hand, and the numerical values in this document are read from the analysis output rather than transcribed, so the report cannot drift from the results it describes.

Command	Produces
python scripts/verify_stage0.py	Solver verification against analytic echo times
python scripts/run_stage3.py	Full-wave B-scans
python scripts/run_stage4.py	Validation gate
python scripts/run_matrix.py	The 192-cell experiment matrix
python scripts/run_stage7.py	Statistical analysis plan
python scripts/run_band_study.py	Bandwidth limitation study
python scripts/tune_feed_taper.py	Antenna feed optimisation
python scripts/measure_gain.py	Far-field gain
python scripts/run_sar.py	Exposure assessment
python scripts/make_figures.py	All figures in this report
python scripts/make_antenna_fab.py	DXF, STEP and drawing
python scripts/make_report.py	This document
python -m pytest tests/	146 tests

Corrections made during the work. Six conclusions recorded earlier in the project were later found to be incorrect and are corrected in the repository documentation: a solver convergence claim that measured operating-system suspension rather than computation; the prescribed remedy for the antenna shortfall; the diagnosis of that shortfall as a cutoff; a fabrication drawing that would have misstated trace width at the connector; a unit test asserting the wrong dimension; and a figure double-counting two cells. Each is documented where it occurred rather than silently revised.

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Appendix A Bill of Materials for a Physical Build

The project is purely virtual by design, and nothing listed here has been bought. This is what a physical build would need, with each specification traced to the simulation that fixed it rather than to a generic shopping list. Cost figures are the proposal's indicative Indian-supplier numbers (Robu, Element14 India, Mouser India, Amazon.in) and **must be re-quoted before purchase**; several are a year old.

Configuration	Approximate total
With a department VNA (recommended)	**INR 23,800**
With the LiteVNA-64 fallback	**INR 32,800**

The one thing the proposal could only guess at — whether the cheap VNA fallbacks actually work — is now measured (Section A.2). The spreadsheet form, *docs/bill-of-materials.csv*, carries all three VNA options, so its column sum (INR 42,700) is not a configuration total.

A.1 The measurement instrument

The VNA decides viability. The proposal's framing stands: *the single question is whether the department owns a VNA reaching 8 GHz or above.*

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
VNA, preferred	1-port S11; sweep 3–11 GHz (antenna is matched 2.48–12.00 GHz); at least 201 points; output 0 dBm; dynamic range 95 dB or better at about 1 kHz IF bandwidth	Department lab VNA, 8–20 GHz	1	0	Check first; decides everything below
VNA, fallback 1	50 kHz – 6.3 GHz	LiteVNA-64	1	9,000	**Works**; measured below
VNA, fallback 2	To 4.4 GHz	NanoVNA V2 Plus4	1	7,000	Works on band; dynamic range untested
Calibration kit	SMA short / open / load, calibrated **at the antenna connector**, not at the VNA port	SMA SOL kit (often bundled with the LiteVNA)	1	1,500–3,000	Everything downstream is referenced to the antenna feed

Where the numbers come from. The matrix was solved over 3–11 GHz at 201 points. The error model assumes a **95 dB** noise floor below the strongest return, which a lab VNA at 1 kHz IF bandwidth meets and a LiteVNA (roughly 70 dB at its top end) does not. **0 dBm** is the adopted transmit power from the exposure assessment: at +10 dBm the 6–11 GHz absorbed power density sat within 1.8× of the ICNIRP public limit; at 0 dBm every limit clears by at least 18×, and the detector has 40.8 dB of margin to spend (Section 7).

A.2 Do the cheap VNAs work? Measured, not guessed

The proposal labelled the LiteVNA “band-limited; degrades resolution” and the NanoVNA “minimum viable, marginal”. Those were guesses made before the matrix was solved. Every solved cell stores the full 3–11 GHz response, so each fallback can be scored on exactly the data it could have acquired, with no new simulation (*scripts/vna_fallback_study.py*, same pipeline and detector as Stage 7).

VNA option	Usable band	Bandwidth	Range res.	AUC	Class separation
Department VNA	3.00–11.0 GHz	8.00 GHz	3.67 mm	1.0000	**40.8 dB**
(first antenna, for scale)	4.35–11.0 GHz	6.65 GHz	4.42 mm	1.0000	17.9 dB

VNA option	Usable band	Bandwidth	Range res.	AUC	Class separation
LiteVNA-64	3.00–6.30 GHz	3.30 GHz	8.91 mm	1.0000	**51.5 dB** at 95 dB dynamic range; **36.4 dB** at 70 dB
NanoVNA V2 Plus4	3.00–4.40 GHz	1.40 GHz	21.0 mm	1.0000	**43.0 dB**, band only, at 95 dB dynamic range

Verdict: both work, and on the band alone the LiteVNA beats the department VNA. Removing the upper band *raises* the separation, 51.5 dB against 40.8, because the frequencies above 6.3 GHz contribute little through 6 mm or more of overlay and mostly add image background the detector does not want; the low band carries the penetration (Section 6). By overlay, the LiteVNA band gives 64.9 / 66.6 / 65.2 / 51.5 dB at 0 / 3 / 6 / 10 mm and the NanoVNA band 54.8 / 57.3 / 60.5 / 45.9 dB. The classes do not overlap at any overlay for either.

Dynamic range, measured separately. The band study keeps the error model's 95 dB noise floor, which the cheap VNAs do not have. Re-scoring the LiteVNA band at 70 dB (*scripts/vna_dynrange_study.py*, 576 realisations) gives AUC 1.0000 and **36.4 dB** separation: 50.2 / 49.1 / 50.0 / 36.4 dB by overlay. The 25 dB worse floor costs 15 dB of margin, nearly all of it at the 10 mm overlay, and what remains is within 5 dB of the department VNA's full-band 40.8. A LiteVNA-64 is therefore a defensible instrument for this experiment, not a compromise. The same test has not been run for the NanoVNA band; its 43.0 dB is the band-only figure at 95 dB and should be read as an upper bound on what that instrument would deliver.

A.3 The antenna

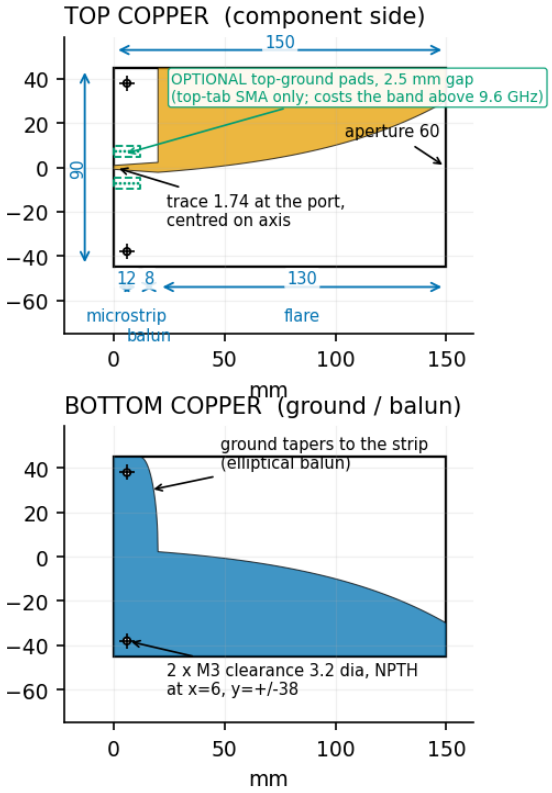
Design files are in *figures/*: *vivaldi.dxf* (4 layers, mm, origin at the feed edge), *vivaldi_antenna.step*, *vivaldi_fab_drawing.pdf* (A4, both faces dimensioned) and *vivaldi_geom.json*. Rebuild with *python scripts/make_antenna_fab.py*.

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Substrate	Rogers RO4350B, 0.762 mm (30 mil), 1 oz Cu, 100 × 150 mm; the antenna is 150 × 90 mm, one board per antenna	RO4350B 0.762 mm	2	3,500	FR4 alternative INR 300; costs 2.4 dB at 9 GHz and 3.6 dB at 11 GHz (Stage 5)
Fabrication	Etch both faces from *vivaldi.dxf*; four 3.2 mm mounting holes at (6, ±38) mm	PCB house, or in-house etch	2	3,000	One spare
SMA connector	**End-launch, 18 GHz rated**; port trace is 1.74 mm (50 ohm)	L-com RFPCB-SMA-F S18G-55 (solder, cut for 0.031 in board) or Southwest 292-04A-9 (solderless clamp, 0.010 in pin)	4	1,600	**Do not use cheap 6 GHz parts.** No extra pads needed; see note
Mounting	M3 nylon standoffs and screws for the four holes	M3 × 10 nylon	8	200	Keeps metal out of the near field

The antenna as designed. Antipodal Vivaldi, growth a·T = 2.0, 60 mm aperture, **tapered feed 4.50 mm at the flare to 1.74 mm at the port**. The taper is what took the match from 4.35–12.00 GHz to **2.48–12.00 GHz**: 99.69% of 3–11 GHz below −10 dB, worst −9.98 dB at one point, median Re(Z_{in}) 45.6 ohm (Section 5). Gain is 7.02 dBi at 3 GHz, 11.9–12.5 dBi across 5–9 GHz and 9.3 dBi at 11 GHz; the 8–10 dBi target is met from about 4 GHz up and **not at 3 GHz**, which is aperture-limited and was shown not fixable by resizing within the board.

Connector note, from the fabrication drawing. Both named parts were checked against their datasheets for the 0.762 mm board. An end-launch lands its pin on the 1.74 mm trace at the board edge and grounds

to the full-width plane on the other face, so **no pads are drawn and none are needed**. The Southwest part's 0.0635 in (1.61 mm) dielectric is within 8% of the trace width, which is why it launches cleanly. The proposal's INR 400 per connector assumes a generic 18 GHz end-launch; the named Southwest part is a premium item at roughly ten times that — price it before choosing. Top-ground-tab connectors are supported only with a 2.5 mm or wider pad gap and cost the band above 9.6 GHz; avoid them.



Fabrication drawing of the antipodal Vivaldi, dimensioned panels only; the full sheet with the connector and pad notes is figures/vivaldi_fab_drawing.pdf. Both faces, 150 × 90 mm on RO4350B 0.762 mm, feed tapered from 4.50 mm at the flare to 1.74 mm at the port.

A.4 Cabling

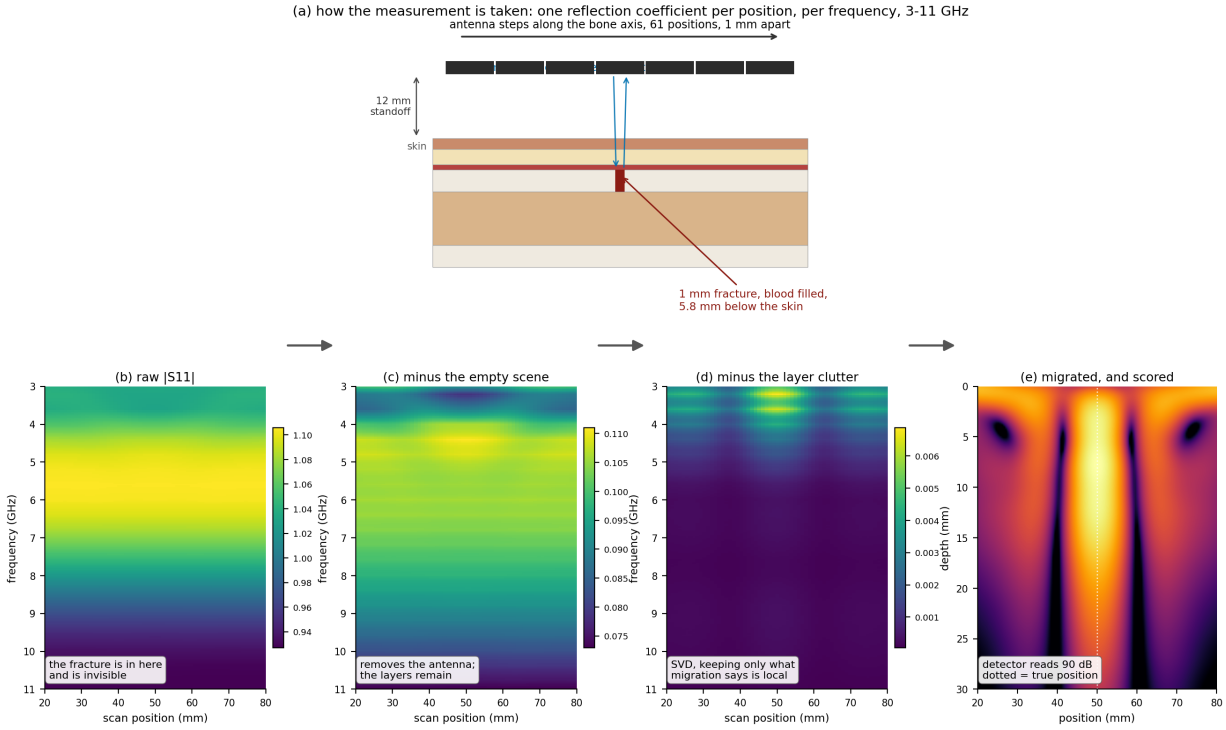
Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Test cable	Phase-stable, SMA–SMA, 18 GHz, 60 cm	Mini-Circuits or Pasternack phase-stable	1	4,500	**Secure it** — flexing changes phase
Port saver	SMA female–female adapter on the VNA port	SMA F–F, 18 GHz	2	500	The VNA connector outlives the project; the saver does not

The error model's cable term is 1 fs RMS delay jitter, i.e. a cable that does not move during a scan. Route it along the stage with slack at a single fixed bend, not dragged.

A.5 The scan stage

Migration sums phase coherently across positions, so position error is phase error: 0.5 mm at 10 GHz in air is about 60 degrees. The error model assumes **100 µm RMS** stage jitter and the result tolerated it; build to that.

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Rail	At least 60 mm of scan travel (the matrix used 61 positions at 1 mm, 20–80 mm); 200 mm gives clearance for the 150 mm antenna	2020 aluminium extrusion, 200 mm	1	2,800	Includes lead screw
Lead screw	8 mm, **anti-backlash nut**	T8 with anti-backlash brass nut	1	(incl.)	Backlash is a systematic position error migration cannot remove
Motor and driver	NEMA 17, A4988, 12 V 3 A supply	Standard 3D-printer parts	1	1,400	1 mm per step command; settle 200 ms before each sweep
Home switch	Mechanical limit switch at one end	Microswitch	1	(incl.)	Repeatable zero; required for empty-scene subtraction to line up
Controller	Serial command: step, settle, trigger	Arduino Nano or ESP32	1	600	The laptop drives the VNA over USB; the controller only moves the stage



The measurement the stage and fixture must reproduce: 61 antenna positions at 1 mm along the bone axis, 12 mm standoff, one S11 sweep per position, then empty-scene subtraction, clutter removal and migration.

A.6 Fixture

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Antenna holder	**12 mm standoff** from aperture to skin, held to **±0.3 mm**	Acrylic or 3D-printed bracket on the stage carriage	1	(in 2,500)	standoff_sigma = 300 µm in the error model
Limb / phantom rest	Holds the specimen with the bone axis parallel to the rail	Acrylic cradle	1	(in 2,500)	Tilt shows up as a sloping skin line
Phantom tank	For casting layered phantoms around the bone	Acrylic box with lid	1	(in 2,500)	Cast overlay slabs separately (next section)
Frame	Mounts rail, holder and rest in one rigid assembly	2020 extrusion and brackets	1	2,500 total	Rigidity is what makes the 100 µm stage spec achievable

A.7 The phantom and specimens

The simulated layer stack is the casting target. Thicknesses are from *bonesim/geometry/phantom.py*; dielectric targets per tissue are in Section 3 (Gabriel four-Cole-Cole).

Layer	Thickness	Note
Skin	2.0 mm	Low oil fraction
Fat	3.0 mm	High oil fraction
Muscle	1.0 mm	High water
Near cortical wall	4.0 mm	The bone specimen itself
Marrow	10.0 mm	The bone specimen itself
Soft-tissue overlay	**0, 3, 6, 10 mm**	The four experimental conditions; cast as separate slabs laid over the skin layer

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Phantom materials	Oil-in-gelatin recipe (proposal 4.3)	DI water 50–70%, porcine gelatin 250 bloom 10–15%, kerosene or safflower oil 15–35% (the tuning knob), surfactant 1–2%, n-propanol 0–5%, p-toluic acid under 0.5%, NaCl 0–1%	–	1,800	Oil fraction sets permittivity 10–50; **measure each batch** with an open-ended coaxial probe on the VNA
Bone specimens	Mid-shaft, about 4 mm cortical wall, about 10 mm medullary cavity	Goat or bovine tibia, fresh from a butcher	6–12	900	Defat mechanically; **do not boil**; store in saline gauze at 4 °C
Fracture fill	Blood-mimicking fluid; haematoma is the real contrast	Saline-based blood mimic	–	(in 1,800)	The simulation used Gabriel blood

Item	Specification fixed by the simulation	Suggested	Qty	INR	Note
Validation target	**6 mm brass rod at 20 mm depth**	Brass rod, 6 mm diameter, 100 mm	1	200	**Stage 4 gate: image this before any bone.** If migration cannot find it, nothing downstream is meaningful
Kerf tools	Kerfs of 0.2–2.0 mm were simulated; the saw sets what is achievable	Jeweller's saw, diamond wafering blade	–	(in 1,200)	Measure the actual kerf; the measurement is the ground truth
Metrology	–	Digital calliper, feeler gauges	–	(in 1,200)	Record every kerf and every cortical thickness as covariates
Consumables	–	Cling film, gloves, saw blades	–	1,200	–

A.8 Compute and software

Nothing to buy. The laptop that ran the analysis runs the acquisition.

Item	Note
bonesim package	Processing chain, detector, statistics — unchanged between simulation and bench
VNA control	scikit-rf and pyvisa for a lab VNA; LiteVNA and NanoVNA have serial protocols with Python libraries
Stage control	Serial to the Arduino or ESP32

The processing chain was built so that it cannot tell the analytic model, the FDTD solver and a real VNA apart: it sees a matrix of complex S11 and nothing else. That is the point at which the simulation hands over to the bench.

A.9 What the build would test that the simulation could not

In order of how much they threaten the result (Section 8):

The error model. 95 dB dynamic range, 100 µm stage jitter, 1 fs cable jitter and ±0.3 mm standoff are assumptions. The 40.8 dB margin is the budget the real bench has to stay inside. This is what a brass rod and a VNA measure on day one.

Real bone. Eight simulated specimens vary only in what the variation model perturbs. Goat tibiae vary in everything.

Fracture position. Every simulated fracture is at x = 50 mm. A real cut is wherever the saw went, which tests localisation rather than separability.

Three dimensions. The bench is 3D for free. The 3D model (*bonesim/geometry/cylindrical.py*) is built and verified but unsolved.